

## Research article

# Business model innovation and firm performance: Evidence from manufacturing SMEs

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## ABSTRACT

In the current volatile business environment, companies are obliged to search for new ways of doing business to remain competitive. Accordingly, firms innovate their business model as it became a promising strategy to achieve sustainable outcomes. However, there is still a need for empirical studies that examine the relationship between business model innovation (BMI) and the performance of small and medium-sized enterprises (SMEs). In this study, we aimed to investigate this relationship by collecting data from 264 manufacturing SMEs through structured questionnaires. We employed partial least square structural equation modeling (PLS-SEM) to analyze the collected data and test the hypotheses. The results indicated that changes in any component of the business model, namely value creation, value proposition, or value capture, had a positive and significant relationship with the performance of manufacturing SMEs. Therefore, by innovating their business models, firms can create more value for their customers while capturing value for themselves. In conclusion, increasing use value or decreasing exchange value with customers will help firms create more value and surpass competitors in the marketplace, while also allowing them to capture more value for themselves.

## 1. Introduction

In the current volatile environment, companies are obliged to rethink and search for unique strategies and innovative ways of doing business to improve their performance [1] and remain competitive [2]. For this purpose, firms often make significant efforts to innovate their products and processes to increase their revenue and maintain and/or improve their profit level [3].

The Organization for Economic Co-operation and Development (OECD) defines innovation broadly as “the introduction of a new or significantly improved product (good or service) or process, new marketing method or a new organizational method in business practice, workplace organization, or external relations” [4]. However, innovations in products and processes are insufficient to compete in the current market where technology development, globalization, ease of access to information, and development in the world economy are high [5,6]. In addition, innovations to improve products and processes are often costly and time-consuming, require high investment in research and development, and require new equipment and even entirely new business units whose future pay-back is uncertain [3,5]. For these purposes, most firms are now turning to Business Model Innovation (BMI) as an alternative or even complement to product or process innovation, as BMI goes beyond just innovating products, services, or technology that can be easily replicated [3,7].

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The business model, which describes how a firm creates, delivers, and captures value [8], should be either slightly modified or completely changed in response to the changing conditions and strengthened over time as the competitive environment evolves [9]. Thus, BMI is defined as “the conscious change of an existing business model or the creation of a new business model that better satisfies the needs of the customer than the existing business model” [10,11]. Currently, interest in BMI, and business models in general, is increasing in both practice and research [12,13]. In a global study conducted by the Economic Intelligence Unit (EIU), the majority of senior managers stated they prefer to innovate business models over product and process innovations [3]. As such, BMI has recently received great attention from scholars and academicians in various fields of study [13,14].

However, empirical research linking BMI to other factors in firms is still scarce and dispersed across different disciplines [12,15]. Most previous studies have primarily focused on defining and explaining the concept and differentiating it from other management concepts [11,16]. Furthermore, empirical studies in this area have yielded conflicting results. While some studies have found a positive relationship between BMI and firm performance [12,17], others have found a negative [18,19], or non-significant relationship between the two [20]. Despite limited research indicating the relationship between BMI and firm performance, the question of whether changing the business model results in a change in firm performance remains unanswered [12,13]. Moreover, answering this question is very important for small and medium-sized enterprises (SMEs) engaged in manufacturing, as they have fewer resources and capabilities than large firms [17]. Therefore, given the importance of SMEs in the global economy, empirical research is needed to ascertain whether BMI activity in manufacturing SMEs is associated with improved performance [12,17]. This could be achieved through large-scale empirical investigations of the relationship between BMI and firm performance, utilizing statistical methodologies that ensure greater generalizability of results, as recommended by scholars [11,21].

Therefore, the objective of the current study is to empirically investigate the relationship between BMI and the performance of manufacturing SMEs. By doing so, first, the study broadens the scope of business model research, which has previously focused on defining and differentiating the concept from other concepts and identifying its drivers and antecedents [8,12,16]. Second, previous studies in the area have relied on qualitative data and are characterized by an explorative approach to gain a first-hand understanding of the concept [11]. Furthermore, some empirical studies have made use of secondary data collected for other purposes [12,22]. In contrast, the current study utilizes statistical methods and quantitative data analysis, thereby providing methodological contributions to the existing literature.

The subsequent sections of this paper are organized as follows. Section 2 provides a review of the empirical literature and hypothesis development. Section 3 outlines the methods and materials used in this study. Section 4 presents the results, while Section 5 discusses the implications of the findings with concluding remarks. Finally, Section 6 discusses the implications, limitations, and future research directions.

## 2. Empirical review and hypothesis development

### 2.1. Business model and business model innovation

The concept of business model has gained widespread acceptance in corporate practice and high visibility in strategic and entrepreneurship research [16,23]. There is no single widely accepted definition of a business model because the literature develops in silos based on the interests of respective researchers [5,16]. Amit and Zott [24] define the business model as “the content, structure, and governance of transactions designed so as to create value through the exploitation of business opportunities.” Chesbrough and Rosenbloom [25] describe the same as “the heuristic logic that connects technical potential with the realization of economic value.” While the concept and definition of a business model differ, various scholars have conceptualized it based on its basic dimensions and agree that its main purpose is to create and deliver value to customers and enterprises themselves [6,8]. In this study, we adopt Teece’s [8] definition of a business model, which portrays it as “the design or architecture of the value creation, delivery, and capture mechanisms of an organization.”

The first basic element of the business model, value creation, refers to the tasks a firm performs to provide an offer to customers by using its resources and capabilities [14]. It also indicates the firm’s ability to acquire resources, such as land, capital, and labor, and transform them into products and services [6]. Value can be created by either increasing the customers’ willingness to pay or decreasing the suppliers’ and partners’ opportunity costs [26]. On the other hand, the value proposition dimension indicates the company’s bundle of products and services that are of value to customers and the way in which they are offered [6,14]. It also explains how a particular firm differentiates itself from its competitors and the reason why customers buy from that firm rather than others [6]. Finally, the value capture dimension defines how value offerings are converted into revenue streams and then captured as profits by firms [14]. According to Chesbrough [5], the value capture dimension is equally critical to a firm’s success because a company that cannot profit from some of its activities cannot sustain those activities over time.

The business model requires constant vigilance because it cannot be static and last forever by efficiently unlocking, capturing, and redistributing the value added by organizations [8,15]. As a result, it must be either slightly modified or completely changed in response to the changing environmental conditions and strengthened over time [9]. Thus, BMI is defined as “the discovery of new or conscious change of enterprise’s existing value creation, delivery, and/or capture mechanism that better satisfies customer needs than the existing business model” [8,10].

The concept of a business model is fundamentally linked to technological innovation. For this purpose, different scholars have put forth various perspectives on the interaction between business models and technology. While some argue that developments in technology facilitate new business models [5,13,27], others argue that the emergence of business models pushed for the development of new technologies [9]. However, they all stated unequivocally that innovation in the business model is distinct from innovation in

technology. According to Baden-Fuller and Haefliger [27], the business model is essentially separable from technological innovation, even though there exists a fundamental linkage between the two. In addition, Chesbrough [5] clearly indicated that innovation in the business model is not the same as innovation in technology. In his words, “better business model often will beat a better technology.” In a later study, Chesbrough [28] pointed out that companies commercialize their technologies through business models. Otherwise, the economic value of technology remains latent until firms introduce it through a business model. Therefore, firms should innovate their business models in order to reap the outcomes of technological innovations.

## 2.2. Firm performance

Currently, firm performance has become a relevant concept and ultimate outcome variable in any management research [29]. Despite its prevalence in the academic literature, there is less consensus about its definition and measurement among researchers [30]. After the seminal work of Venkatraman and Ramanujam [31], scholars have started to recognize and measure firm performance as a multidimensional construct [29,32]. Nevertheless, while acknowledging its multidimensionality, several other researchers measure firm performance as unidimensional [32].

One can measure firm performance objectively (through accounting and financial measures) or subjectively (via survey-based self-reports) [30]. While objective measures are preferable over subjective measures, objective indicators on the performance of SMEs are difficult to collect [30,32] and, in some cases, impossible to collect [31]. Singh et al. [30] also indicated that obtaining consistent and comparable data in the objective measures of performance for the entire sample under investigation is extremely difficult. Furthermore, most owners are not willing to disclose information regarding objective performance measures [33]. For these purposes, several studies adopt subjective perception-based measures of firm performance [30,32,33]. However, whether objective or subjective measures of performance are used is up to the researcher [30].

## 2.3. The relationship between BMI and firm performance

There is increasing consensus among scholars regarding the relationship between BMI and firm performance [16]. According to Pohle and Chapman [34], BMI is responsible for a significantly larger improvement in the firm's performance than innovation in product and process. In addition, the authors stressed that BMI is often positively related to reduced costs and, thus, enhanced profits for manufacturers. Teece [8] also indicated that a well-designed business model that includes all aspects (components), along with the implementation and refinement of commercially feasible revenue and cost architectures, is critical to enterprise success. Furthermore, Cucculelli and Bettinelli [22] pointed out that modifying a firm's business model in an innovative way is positively associated with venture performance. In the same way, scholars who conceptualized business models based on their components reported a positive relationship with firm performance. For instance, Clauss et al. [35] found that innovation in value creation increases firm performance by delivering greater economic results. Similarly, a study by Chen et al. [17] in manufacturing SMEs indicated that value creation innovation is positively and significantly related to SMEs' performance. Therefore,

**H1.** Value creation innovation is positively related to the performance of manufacturing SMEs.

Innovation in the value proposition helps firms to extend their product and service portfolios and address new market needs, which in turn is highly associated with their performance [35]. According to Clauss et al. [35], a change in value proposition will result in a change in customer offerings, which includes product/service, target markets, and delivery channels. Clauss et al. [36] also indicated that value proposition innovation has a positive relationship with firm performance. Furthermore, Chen et al. [17] found that value proposition innovation is positively and significantly associated with manufacturing SMEs' performance. Therefore,

**H2.** Value proposition innovation is positively associated with the performance of manufacturing SMEs.

Renewing the value capture mechanism can help firms replace less profitable revenue sources and improve their potential profits [26]. Casadesus-Masanell and Zhu [10] also indicated that innovations in value capture help firms to realize new revenue streams in addition to existing ones, or to substitute less profitable ones compared to other revenue capture mechanisms, which improves the prospects for future revenue. The authors also emphasized that value capture innovation can reduce inefficiencies and, as a result, improve firms' performance. Therefore,

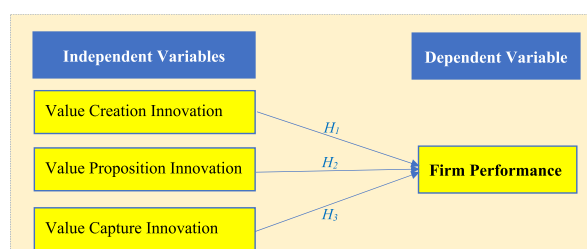


Fig. 1. Conceptual framework.

### H3. Value capture innovation is positively related to the performance of manufacturing SMEs.

The overall structure of the relationship between the study variables and firm performance is indicated in Fig. 1.

## 3. Methods and materials

### 3.1. Sample and data

The aim of this study is to investigate the relationship between BMI and manufacturing SMEs' performance. To achieve this, we used primary data collected from SMEs located in Addis Ababa, Ethiopia. At the time of survey (between May and October 2022), there were 1201 small and 602 medium-sized manufacturing enterprises. We proportionally and randomly selected 318 manufacturing SMEs to include in the study sample to ensure that each sector received fair representation. We excluded startups from the study as they don't have a fully developed business model and lack adequate capital to move on to the next phase [33]. We collected 276 questionnaires, with eight having limited data. Therefore, we analyzed 264 correct responses. Out of correct responses, 173 were small and 91 were medium-sized enterprises. In Ethiopia, small enterprises have a total capital ranging from Birr 100,001<sup>†</sup> to Birr 1,500,000 and employ 6 to 30 workers, including the owner, family members, and other employees. Medium enterprises have a total capital (excluding building) ranging from Birr 1,500,001 to Birr 20,000,000 and employ 31 to 100 workers, including the owner, family members, and other employees (Federal Urban Job Creation and Food Security Agency [FUJCFSA], 2019). The data also showed that 96 enterprises were engaged in wood and steel work, 91 in textile and leather manufacturing, 48 in agro-processing, and the remaining 29 were engaged in the manufacturing of chemicals and industrial inputs. The classification of manufacturing SMEs into sub-sectors in this study was based on the Ethiopian government's classification system.

We collected data from owners/managers of SMEs using structured questionnaires. Before collecting data, we obtained informed consent from all participants involved in our research. The consent form outlined the purpose of the study and the fact that all data collected would be kept confidential and used solely for academic purposes. We administered the questionnaire on-site to owners/managers of each enterprise. The questionnaire had four sections: background information, BMI, environmental dynamism, and firm performance. The background information section included gender, the respondents' role in the enterprise, educational level, enterprise age, manufacturing sub-sector, and the number of employees. We collected BMI data using an instrument developed by Clauss [14]. The instrument has 30 Likert-scaled items in which 10 items measure value creation innovation, 12 measure value proposition innovation, and the remaining 8 measure value capture innovation (see Appendix 1). Each item ranges from a scale of 1 (strongly disagree) to 5 (strongly agree). The first-order constructs in the measurement scale consist of new capabilities, which can be developed through training and knowledge integration; new technology, which is related to equipment used to carry out BMI; new processes, which are concerned with how activities are connected; new partnerships, which represent external resources available to the firm; new offerings, which justify the firm's product offerings; new markets, which identify the customer group in which the product is offered; new channels, which are concerned with the delivery of value to customers; new relationships, which indicate the firm's ability to build or establish a relationship with customers; new revenue models, which are concerned with transforming revenue models to encourage customers to pay for the value proposition; and new cost structures, which are related to the direct and indirect costs of running the business [14]. We collected data on performance of SMEs by using financial and marketing indicators such as sales growth, profit growth, market share, speed to market, net income, and return on investment, as recommended by Venkatraman and Ramanujam [31], and we measured them using a Likert-type scale, with response options ranging from 1 (strongly disagree) to 5 (strongly agree).

### 3.2. Measurement of variables

The study primarily includes two variables: BMI and firm performance, which are predictor and outcome variables, respectively. BMI is concerned with changes in one or more components of the business model, such as value creation, value proposition, and value capture [14]. Firm performance encompasses both financial and non-financial measures [31].

Previous studies have used different measurement scales to measure BMI. For instance, Velu [37] used diversification/product launch and external funding as two indicators of BMI while Anwar [38] utilized six items developed by Karimi and Walter that focused on different aspects of a firm's innovativeness, such as product and service, delivery, process, and structure. On the other hand, Clauss [14] developed and validated a BMI measurement scale, which Bouwman et al. [12] identified as a valuable contribution to measuring BMI. As a result, we measured BMI by assessing the changes introduced in the three dimensions of the business model: value creation, value proposition, and value capture. To capture the multi-dimensional nature of BMI, we used a reflective-formative measurement model and estimated hierarchical latent variables: value creation innovation, value proposition innovation, and value capture innovation using a two-stage approach in PLS-SEM, which has advantages over other methods [39].

We measured firm performance using Venkatraman and Ramanujam's performance measurement scale [31]. Initially, the authors identified three dimensions of business performance, namely financial, operational, and market performance. However, they later combined the market and financial performance dimensions into one, resulting in a two-dimensional framework of business performance—financial and operational—that has become increasingly prevalent in recent empirical research [12,32,33]. We used

<sup>†</sup> 1 USD = 54.35 Birr as of April 24, 2023.

subjective data to measure the performance of manufacturing SMEs since objective data on their performance is difficult to obtain as they do not publish their financial results [32]. Furthermore, owners are typically unwilling to voluntarily disclose their business's financial data to outsiders [30,33]. Therefore, we ask owners/managers of enterprises to rate their company's performance against primary competitors within the industry during the last two years.

Internal and external conditions of a firm may affect the relationship between BMI and firm performance [26]. To reduce potential confounding effects caused by omitted variables, we incorporated three control variables into our model: firm age, size, and environmental dynamism. Semrau et al. [40] also recommend controlling the firm's age and size for better results. We measured firm age based on the logarithm of the number of years since the establishment, firm size with the logarithm of the number of employees, and we measured environmental dynamism by a measurement scale developed by Ting et al. [41], which includes changes in technology; variations in customer preferences, product demand, and competitors; and fluctuations in the supply of materials.

#### 4. Results

From the total of 318 questionnaires distributed to manufacturing SMEs, 276 have been returned, resulting in an 86.8% response rate. We excluded 12 questionnaires from the analysis because eight were incorrectly filled and four were filled by startups that were not part of our study. Therefore, we analyzed 264 correct responses.

##### 4.1. Descriptive results

We used descriptive statistics to summarize the demographics of our sample. Of 264 respondents, 65.2% were male and 34.8% were female. The age distribution of data indicated that 38.3% of the sample were between the ages of 18 and 30, 42.8% were between the ages of 31 and 40, 11.4% were between 41 and 50, and the remaining 7.6% were above 50 years old. Of the respondents who participated in the survey, 76.1% were owners, 19.3% were managers, and the remaining 4.5% were representatives of the owners/managers. The age distribution of firms showed that 46.6% had been in business for 3–8 years, 40.9% for 9–14 years, and the remaining 12.5% for 15 years or more (see Table 1).

The new capabilities had the highest mean score ( $\bar{X} = 4.120$ ,  $S = 0.594$ ) compared to other constructs, followed by the new cost structure ( $\bar{X} = 4.039$ ,  $S = 0.605$ ). This indicates that, compared to others, manufacturing SMEs train their employees to help them catch up-to-date knowledge, adapt to changing markets, regularly reflect on price-quantity strategy, seek opportunities to save manufacturing costs, constantly examine market prices and act accordingly, and use opportunities of price differentiation [14]. The new channels had the lowest mean score ( $\bar{X} = 2.976$ ,  $S = 0.745$ ), followed by the new revenue models ( $\bar{X} = 3.171$ ,  $S = 0.690$ ), compared with others (see Table 2). This indicates that, relative to others, manufacturing SMEs do not regularly change their

**Table 1**  
Background information.

| Items                            | Distribution                 | Frequency (f) | Percentage (%) |
|----------------------------------|------------------------------|---------------|----------------|
| Gender of Respondents            | Male                         | 172           | 65.2           |
|                                  | Female                       | 92            | 34.8           |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
| Age of Respondents               | 18–30                        | 101           | 38.3           |
|                                  | 31–40                        | 113           | 42.8           |
|                                  | 41–50                        | 30            | 11.4           |
|                                  | 51 and above                 | 20            | 7.6            |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
|                                  |                              |               |                |
| Responsibility in Enterprise     | Owner                        | 201           | 76.1           |
|                                  | Manager                      | 51            | 19.3           |
|                                  | Representative               | 12            | 4.5            |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
| Educational Level of Respondents | Highschool and below         | 104           | 39.4           |
|                                  | Associate's Degree           | 90            | 34.1           |
|                                  | Bachelor's Degree            | 67            | 25.4           |
|                                  | Master's Degree              | 3             | 1.1            |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
| Age of Firm in Years             | 3–8 years                    | 123           | 46.6           |
|                                  | 9–14 years                   | 108           | 40.9           |
|                                  | 15 & above years             | 33            | 12.5           |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
|                                  |                              |               |                |
| Sub-sector of Manufacturing      | Wood and Metal Work          | 96            | 36.4           |
|                                  | Textile and Leather          | 91            | 34.5           |
|                                  | Agro-processing              | 48            | 18.2           |
|                                  | Chemical & Industrial Inputs | 29            | 11.0           |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |
| Number of Employees              | 6–30                         | 173           | 65.5           |
|                                  | 31–100                       | 91            | 34.5           |
|                                  | <b>Total</b>                 | <b>264</b>    | <b>100.0</b>   |

**Table 2**

Means, standard deviation, and correlation matrix.

|                     |    | Mean  | Standard Deviation | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     |
|---------------------|----|-------|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| New Capabilities    | 1  | 4.120 | 0.594              |        |        |        |        |        |        |        |        |        |        |
| New Technology      | 2  | 3.620 | 0.670              | .702** |        |        |        |        |        |        |        |        |        |
| New Partnerships    | 3  | 3.513 | 0.649              | .551** | .651** |        |        |        |        |        |        |        |        |
| New Processes       | 4  | 4.003 | 0.583              | .501** | .569** | .540** |        |        |        |        |        |        |        |
| New Offering        | 5  | 3.553 | 0.704              | .621** | .603** | .606** | .541** |        |        |        |        |        |        |
| New Markets         | 6  | 3.479 | 0.706              | .509** | .534** | .513** | .492** | .728** |        |        |        |        |        |
| New Channels        | 7  | 2.976 | 0.745              | .527** | .446** | .485** | .479** | .646** | .631** |        |        |        |        |
| New Relations       | 8  | 3.788 | 0.631              | .366** | .406** | .401** | .490** | .483** | .473** | .546** |        |        |        |
| New Revenue Models  | 9  | 3.171 | 0.690              | .482** | .494** | .506** | .431** | .542** | .515** | .612** | .536** |        |        |
| New Cost structures | 10 | 4.039 | 0.605              | .325** | .308** | .299** | .485** | .375** | .439** | .416** | .541** | .571** |        |
| Firm Performance    | 11 | 3.103 | 0.649              | .519** | .485** | .473** | .456** | .511** | .478** | .494** | .475** | .518** | .481** |

\*\*. Correlation is significant at the 0.01 level (2-tailed).

distribution channels, have not benefited from channels change, have not recently developed new revenue opportunities, do not offer integrated services, do not practice recurring revenue models, and rely heavily on existing revenue sources [33].

#### 4.2. Correlation matrix

We used Pearson Correlation to indicate the association between variables. According to Norman [42], parametric analysis methods can be used with Likert-scaled measurements without fear of “coming to the wrong conclusion.” As shown in Table 2, all the constructs were positively and significantly correlated with firm performance. We did not observe any multicollinearity issue in our analysis since all the constructs had correlation values below 0.80 [43].

#### 4.3. Common method bias

Because we collected both independent and dependent data from the same sources at the same time, common method bias (CMB) may threaten the results [44]. The researchers recommend two methods: procedural and statistical, for mitigating CMB [45,46]. We implemented several procedural remedies to minimize the occurrence of CMB before conducting the survey. We assured respondents that their anonymity would be protected, and we attempted to reduce participants' evaluation apprehension over their responses by informing them that this was just a survey. According to Podsakoff et al. [45], the measures taken before the experiment can reduce the likelihood and severity of CMB.

Furthermore, we performed Harman's single-factor method using principal component analysis. According to Tehseen et al. [46], this method loads all items from each construct into an exploratory factor analysis to see whether one single factor emerges or one general factor accounts for a majority of the covariance among the measures. While some argue that Harman's single-factor method is inadequate for assessing CMB [45], other recent studies indicate that it is an easy and meaningful tool to assess CMB [46,47]. The results revealed that the total variance extracted by a single factor for the sample was 39.5% (see Appendix 2). Therefore, CMB was not a pervasive issue in this study, as it was below 50% [46].

#### 4.4. Validity and reliability

We assessed the internal consistency reliability of measurement models using Cronbach's alpha, which is commonly used to measure internal consistency with a threshold value of 0.70 [48]. All the values are within the specified threshold. Internal consistency can better be measured by composite reliability than Cronbach's alpha [48,49]. As a result, we measured composite reliability for our measurement model, and, as indicated in Table 3, all the values were higher than the threshold value of 0.70. According to Pesämaa et al. [47], both Cronbach's alpha and composite reliability should exceed 0.70 to be considered acceptable. Furthermore, rho\_A—which is recommended as an accurate and approximate measure of reliability [48]—was found to be within the acceptable range [50], as shown in Table 3.

In addition, we assessed the convergent validity of the construct to investigate its ability to effectively explain the variance of its items. According to Ringle et al. [49], the average variance extracted (AVE) metric can be used to measure the construct's convergent validity, with values of 0.50 and higher being considered acceptable. Therefore, as shown in Table 3, all the AVE values in our measurement model were greater than 0.50 and were acceptable. We also examined the model for discriminant validity using the heterotrait-monotrait (HTMT) ratio of correlations. As shown in Table 4, all the HTMT ratio values in our measurement model were less than or equal to 0.85, which is a strong threshold value [51].

Furthermore, we tested the normality of the data using the Shapiro-Wilk test of normality and the results indicated that none of the constructs followed a normal distribution, with p-values less than 0.05 (see Appendix 3). According to Field [52], if the significance test is greater than 0.05, the distribution is normal; if it is less than 0.05, it deviates significantly from a normal distribution. However, since PLS-SEM is a non-parametric method that does not require distributional assumptions, the lack of normality did not pose a problem [43,53].

#### 4.5. Path model analysis

We used Partial Least Square Structural Equation Model (PLS-SEM) using SmartPLS v.3 to analyze the data and test the hypothesis. We chose the PLS-SEM approach over the CB-SEM approach for several reasons. First, the PLS-SEM approach allows the estimation of very complex models with many constructs and subsequent analysis of latent variables [48]. Second, it places fewer restrictions on data distribution and normality [48,53]. And third, it can extensively estimate interactions among latent predictor variable indicators [43, 48].

##### 4.5.1. Measurement model assessment

The measurement model was used to establish the relationship between the constructs and the measured variables. According to Hair et al. [48], the first step in assessing a reflective measurement model is to examine indicator loadings. Indicator loadings above 0.708 are recommended since they indicate that the construct explains more than 50% of the indicator's variance [54]. Our assessment of the indicator loadings for the reflective measurement models showed that all outer loadings had values greater than the recommended threshold of 0.708 [48], as shown in Table 3. Additionally, we evaluated the reliability and validity of construct measures as per Ali et al. [53]. The assessment indicated that all results met the specified criteria and were acceptable.



**Table 3**

Assessment of Reflective Measurement Model for first order constructs.

| First Order Constructs    | Indicators | Factor Loadings | $\alpha$ | rho_A | CR    | AVE   | VIF   |
|---------------------------|------------|-----------------|----------|-------|-------|-------|-------|
| New Capabilities (CAP)    | CAP1       | 0.838           | 0.819    | 0.821 | 0.892 | 0.734 | 2.094 |
|                           | CAP2       | 0.888           |          |       |       |       |       |
|                           | CAP3       | 0.842           |          |       |       |       |       |
| New Technology (TEC)      | TEC1       | 0.834           | 0.825    | 0.828 | 0.896 | 0.742 | 2.603 |
|                           | TEC2       | 0.900           |          |       |       |       |       |
|                           | TEC3       | 0.848           |          |       |       |       |       |
| New Partnerships (PAR)    | PAR1       | 0.793           | 0.873    | 0.874 | 0.914 | 0.726 | 1.899 |
|                           | PAR2       | 0.886           |          |       |       |       |       |
|                           | PAR3       | 0.861           |          |       |       |       |       |
|                           | PAR4       | 0.865           |          |       |       |       |       |
| New Processes (PRO)       | PRO1       | 0.822           | 0.810    | 0.812 | 0.887 | 0.724 | 1.623 |
|                           | PRO2       | 0.880           |          |       |       |       |       |
|                           | PRO3       | 0.850           |          |       |       |       |       |
| New Offerings (OFF)       | OFF1       | 0.832           | 0.853    | 0.854 | 0.911 | 0.774 | 2.487 |
|                           | OFF2       | 0.920           |          |       |       |       |       |
|                           | OFF3       | 0.885           |          |       |       |       |       |
| New Markets (MAR)         | MAR1       | 0.874           | 0.867    | 0.869 | 0.918 | 0.790 | 2.392 |
|                           | MAR2       | 0.923           |          |       |       |       |       |
|                           | MAR3       | 0.869           |          |       |       |       |       |
| New Channels (CHA)        | CHA1       | 0.886           | 0.882    | 0.883 | 0.927 | 0.809 | 2.106 |
|                           | CHA2       | 0.930           |          |       |       |       |       |
|                           | CHA3       | 0.882           |          |       |       |       |       |
| New Relations (REL)       | REL1       | 0.856           | 0.823    | 0.826 | 0.894 | 0.738 | 1.518 |
|                           | REL2       | 0.869           |          |       |       |       |       |
|                           | REL3       | 0.852           |          |       |       |       |       |
| New Revenue Models (REV)  | REV1       | 0.782           | 0.817    | 0.820 | 0.880 | 0.649 | 1.498 |
|                           | REV2       | 0.873           |          |       |       |       |       |
|                           | REV3       | 0.833           |          |       |       |       |       |
|                           | REV4       | 0.725           |          |       |       |       |       |
| New Cost Structures (COS) | COS1       | 0.803           | 0.845    | 0.847 | 0.896 | 0.684 | 1.498 |
|                           | COS2       | 0.828           |          |       |       |       |       |
|                           | COS3       | 0.877           |          |       |       |       |       |
|                           | COS4       | 0.796           |          |       |       |       |       |
| Firm Performance (FP)     | FP1        | 0.744           | 0.877    | 0.879 | 0.907 | 0.620 | N/A   |
|                           | FP2        | 0.757           |          |       |       |       |       |
|                           | FP3        | 0.840           |          |       |       |       |       |
|                           | FP4        | 0.817           |          |       |       |       |       |
|                           | FP5        | 0.831           |          |       |       |       |       |
|                           | FP6        | 0.727           |          |       |       |       |       |

Note:  $\alpha$  = Cronbach's Alpha; rho\_A = Reliability Coefficient; CR = Composite Reliability; AVE = Average Value Extracted; VIF = Variance Inflation Factor.

**Table 4**

Discriminant Validity of Constructs using HTMT Ratio.

|                           | FP    | CAP   | CHA   | COS   | MAR   | OFF   | PAR   | PRO   | REL   | REV   | TEC |
|---------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----|
| New Capabilities (CAP)    | 0.623 |       |       |       |       |       |       |       |       |       |     |
| New Channels (CHA)        | 0.565 | 0.620 |       |       |       |       |       |       |       |       |     |
| New Cost Structures (COS) | 0.562 | 0.390 | 0.481 |       |       |       |       |       |       |       |     |
| New Markets (MAR)         | 0.555 | 0.606 | 0.723 | 0.510 |       |       |       |       |       |       |     |
| New Offerings (OFF)       | 0.598 | 0.743 | 0.747 | 0.445 | 0.848 |       |       |       |       |       |     |
| New Partnerships (PAR)    | 0.535 | 0.651 | 0.553 | 0.346 | 0.591 | 0.705 |       |       |       |       |     |
| New Processes (PRO)       | 0.543 | 0.613 | 0.566 | 0.588 | 0.587 | 0.653 | 0.641 |       |       |       |     |
| New Relations (REL)       | 0.566 | 0.446 | 0.642 | 0.649 | 0.56  | 0.584 | 0.476 | 0.602 |       |       |     |
| New Revenue Models (REV)  | 0.620 | 0.595 | 0.718 | 0.693 | 0.613 | 0.652 | 0.603 | 0.534 | 0.660 |       |     |
| New Technology (TEC)      | 0.578 | 0.850 | 0.525 | 0.374 | 0.633 | 0.718 | 0.761 | 0.692 | 0.498 | 0.604 |     |

Note: HTMT = heterotrait-monotrait ratio of correlations.

#### 4.5.2. Structural model assessment

After confirming that the measurement model assessment met the relevant criteria, we proceeded to evaluate the structural model results. We assessed the structural model in PLS-SEM for statistical relevance using the coefficient of determination ( $R^2$ ), effect size ( $f^2$ ), Stone-Geisser's predictive relevance ( $Q^2$ ) through the blindfolding procedure, and the statistical significance, after checking collinearity among variables. As shown in Table 5, the value of  $R^2$  was 0.448, indicating that all variables together predict 44.8% of the variation in the performance of manufacturing SMEs. This result is considered satisfactory since Falk and Miller [55] suggest an  $R^2$  value of 0.1 or higher as adequate for explaining the variance of a particular endogenous construct. The values of  $f^2$ , presented in



**Table 5**  
Results of  $R^2$  and  $Q^2$ .

|                  | $R^2$ | $R^2$ Adjusted | SSO      | SSE      | $Q^2 = (1 - SSE/SSO)$ | SRMR Estimated Model |
|------------------|-------|----------------|----------|----------|-----------------------|----------------------|
| Firm Performance | 0.448 | 0.442          | 1584.000 | 1157.962 | 0.269                 | 0.056                |

Note:  $R^2$  = Coefficient of determination,  $Q^2$  = Predictive Relevance, SSO = Sum of Squares Observations, SSE = Sum of Squares Errors, SRMR = Standardized Root Mean Square Residual.

**Table 6**  
Path coefficients and their significance value.

|                         | Path Coefficients | T Statistics | P Value | $f^2$ |
|-------------------------|-------------------|--------------|---------|-------|
| VCRI → Firm Performance | 0.291             | 4.322        | 0.000   | 0.064 |
| CPRI → Firm Performance | 0.216             | 2.505        | 0.013   | 0.030 |
| VCAI → Firm Performance | 0.255             | 3.789        | 0.000   | 0.072 |

Note: VCRI = Value Creation Innovation, the process of changing the tasks that a firm performs to provide an offer to customers; VPRI = Value Proposition Innovation, the process of innovating the company's bundle of products and services and the way in which they are offered to customers; VCAI = Value Capture Innovation, the process of innovating how value offerings are converted into revenue streams and then captured as profits by firms.

Table 6, indicated a medium effect size [49]. Moreover, the predictive relevance,  $Q^2$ , which measures whether the model has predictive relevance or not [48], was good because its value was greater than 0. The path coefficients were acceptable because they fell between  $-1$  and  $+1$  and were statistically significant at the 5% level [54]. We further assessed the model fit using the Standardized Root Mean Square Residual (SRMR), a measure of the difference between the observed correlation and model-implied correlation matrix, and determined that the estimated SRMR value, 0.056, was below the recommended threshold of 0.08 [56]. In general, our results suggested that the structural model was statistically significant and had a satisfactory fit to the data.

#### 4.5.3. Structural model analysis

The structural model displays the relationship between constructs and dependent variables. Hair et al. [48] recommend examining the collinearity before assessing structural relationships to ensure unbiased regression results. Therefore, we first examined collinearity using VIF values. The results, as presented in Table 3, showed that all VIF values were less than 3, indicating that our model was free from any possible collinearity issues [57].

We tested the effect of control variables on endogenous constructs before analyzing our hypothesized variables by including only control variables in the model. While environmental dynamism had a significant effect on company performance, the age and size of a company had no significant effect. The value of adjusted  $R^2$  improved after including control variables in our model.

All the study variables, namely value creation innovation (VCRI), value proposition innovation (VPRI), and value capture innovation (VCAI), were found to have a significant relationship with firm performance. Specifically, H1 evaluated whether VCRI had a significant relationship with SMEs' performance. The results showed that VCRI had a positive and significant relationship with SMEs' performance, supporting H1. These results indicate that when there is a one-unit standard deviation increase in VCRI, manufacturing SMEs' performance also increases by 0.291 standard deviation units. H2 evaluated whether VPRI had a significant association with manufacturing SMEs' performance. The results revealed that VPRI had a positive and significant relationship with the performance of manufacturing SMEs. Therefore, H2 was supported, and a one-unit standard deviation increase in VPRI would lead to 0.216 standard deviation unit increase in the performance of manufacturing SMEs. Finally, H3 tested whether VCAI had a significant relationship with SMEs' performance. The results revealed that VCAI had a significant and positive relationship with SMEs' performance. As a result, H3 was supported, and a one-unit standard deviation change in VCAI would result in 0.255 standard deviation unit change in manufacturing SMEs' performance (see Table 6).

## 5. Discussion and conclusion

The purpose of this study was to investigate the relationship between BMI and manufacturing SMEs' performance. We conceptualized BMI as any changes made to the dimensions of the business model—value creation, value proposition, and value capture.

The structural model indicated that the first hypothesis stating that “value creation innovation is positively related to the performance of manufacturing SMEs” is supported as it has a positive and significant path coefficient. This finding is consistent with a previous study by Chen et al. [17]. According to Priem [58], firms can create value by either increasing the use value or decreasing the exchange value from the customer's perspective, as both increase customer surplus. In addition, Al-Debei and Avison [59] indicate that innovation in the value creation dimension can be achieved through resource configuration, which reflects a firm's ability to integrate various assets in a way that offers a valuable bundle of products and services.

The second hypothesis stating that “value proposition innovation is positively associated with the performance of manufacturing SMEs” is supported because it has a positive and significant path coefficient. This finding is consistent with the previous research by Clauss et al. [35], who found a positive association between value proposition innovation and firm performance. Thus, innovation in value proposition helps firms to attract and retain a large portion of their customer base [59].

Finally, the third hypothesis stating that “value capture innovation is positively related to the performance of manufacturing SMEs” is also supported as it has a positive and significant path coefficient. This result is consistent with the findings of studies by Zott and Amit [26] and Casadesus-Masanell and Zhu [10], both of which indicated that creating new revenue model benefits firms. However, this finding contradicts the findings of Chen et al. [17] and Clauss et al. [35], who both found that value capture innovation is negatively related to firm performance. But the study by Chen et al. [17] indicated that the relationship between value capture innovation and the growth of SMEs is insignificant. To capture value from their innovations, enterprises need to figure out a way to surpass their competitors in the marketplace [25].

Compared with larger firms, SMEs typically have limited financial and non-financial resources, smaller or non-existent R&D facilities, less technical capabilities, difficulty in hiring multi-skilled labor, and a less structured approach to innovation [12,17]. Despite these limitations, if SMEs can find a way to innovate their business model, they can compensate for these difficulties by relying on their strengths that come from their size, such as change receptiveness, less bureaucratic procedures, flexible structures, and high adaptability [60]. To innovate their business model, SMEs can modify a single business element, such as value creation, value proposition, or value capture innovation; change two or more components simultaneously; or change the interaction between elements of the business model [3].

## 6. Implications, limitations, and future research directions

Our study contributes to the business model and BMI literature in several ways. First, we provide theoretical and empirical reasoning for considering BMI as a configuration of different dimensions. Most previous researchers have considered BMI as an aggregate construct [12,17], viewed BMI as the sole change in the value creation dimension of the business model [12], and relied only on proxy measures to measure BMI [11]. In contrast, our study distinguishes three dimensions of the business model, namely value creation, value proposition, and value capture, and investigates the relationship between change in each of the dimensions and SMEs' performance. Second, this study offers more conclusive and rigorous evidence of the relationship between BMI and manufacturing SMEs' performance. Prior researches in the area have been criticized for being inconclusive, lacking rigor, and not being empirical [17].

Furthermore, our study has significant implications for the managerial practices of manufacturing SMEs. Specifically, the study suggests that focusing on customer value propositions instead of just products or services, investing in innovative resources, creating effective partnerships with other strategic parties in the business environment, embracing a distinct market positioning unlike competitors, enticing customers with additional incentives through pricing strategies, and monitoring changing trends in the business environment can help firms obtain early insights into industry developments.

Despite its significant contribution to both theory and practice, the study is not free from some limitations. First, the study relied on subjective measures of firm performance instead of objective ones, which are generally preferred. However, obtaining objective measures is difficult or even impossible for some measures [30,31]. Furthermore, SMEs are not legally required to publish their financial performance, and even if they do, the data may be biased due to the lack of an appropriate auditing system [32,33]. As a result, we relied on subjective data to measure SMEs' performance. Second, our study depended on cross-sectional data obtained from a single informant at the same time for predictor and outcome variables. This may result in common method bias [44]. To minimize this effect, we first used some procedural remedies and then we checked for the availability of this potential problem with Harman's single-factor test using principal component analysis. The result indicated that common method bias did not pose a problem because its value was below 50% [61]. However, future researchers may use a longitudinal research design or collect predictor and outcome variables at different time periods. Third, we physically provided questionnaires to owners/managers of each enterprise. Even though this method of data collection may result in a high response rate, it may compromise the results due to a lack of anonymity. Therefore, future studies ought to use a survey instrument that best preserves the anonymity of respondents. Fourth, there may be circumstances in which the relationship between BMI and firm performance can be strengthened by including contingency variables [13,33]. Therefore, we suggest that future research include the mediating or moderating factors when studying the link between BMI and firm performance.

## Author contribution statement

Natnael Salfore: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Wrote the paper.

Matiwos Ensermu: Zerihun Kinde: Conceived and designed the experiments; Analyzed and interpreted the data; Contributed reagents, materials, analysis tools or data; Wrote the paper.

## Data availability statement

Data included in article/supp. material/referenced in article.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2023.e16384>.

## Appendices.

### Appendix 1

#### Measurement Scale

##### 1) Business Model Innovation (Clauss, 2017).

##### A) Value Creation Innovation.

##### New capability:

- Our employees constantly receive training in order to develop new competencies.
- Relative to our direct competitors, our employees have very up-to-date knowledge and capabilities.
- We constantly reflect on which new competencies need to be established in order to adapt to changing market requirements.

##### New Technology:

- We keep the technical resources of our company up-to-date.
- Relative to our competitors our technical equipment is very innovative.
- We regularly utilize new technical opportunities in order to extend our product and service portfolio.

##### New Partnership:

- We are constantly searching for new collaboration partners
- We regularly utilize opportunities that arise from the integration of new partners into our processes
- We regularly evaluate the potential benefits of outsourcing
- New collaboration partners regularly help us to further develop our business model.

##### New relationships:

- We were recently able to significantly improve our internal processes.
- We utilize innovative procedures and processes during the manufacturing of our products
- Our existing processes are regularly assessed and significantly changed if needed

##### B) Value Proposition Innovation.

##### New offerings:

- We regularly address new, unmet customer needs
- Our products or services are very innovative in relation to our competitors
- Our products or services regularly solve customer needs, which were not solved by competitors

##### New markets:

- We regularly take opportunities that arise in new or growing markets
- We regularly address new, un-served market segments
- We are constantly seeking new customer segments and markets for our products and services

##### New channels:

- We regularly utilize new distribution channels for our products and services

- Constant changes in our channels have led to improved efficiency of our channel functions
- We consistently change our portfolio of distribution channels

#### New Relations:

- We consistently change our portfolio of distribution channels
- We try to increase customer retention through new service offerings
- We emphasize innovative/modern actions to increase customer retention
- We recently took many actions in order to strengthen customer relationships

#### C) Value capture Innovation.

##### New revenue model:

- We recently developed new revenue opportunities (e.g. additional sales, cross-selling)
- We increasingly offer integrated services (e.g. maintenance contracts) in order to realize long-term financial returns
- We recently complemented or replaced one-time transaction revenues with long-term recurring revenue models (e.g. Leasing).
- We do not rely on the durability of our existing revenue sources.

##### New cost structure:

- We regularly reflect on our price-quantity strategy.
- We actively seek opportunities to save manufacturing costs
- Our production costs are constantly examined and necessarily amended according to market prices.
- We regularly utilize opportunities that arise through price differentiation

#### 2) Firm Performance (Venkatraman and Ramanujam, 1986; Pucci et al., 2017; Latifi et al., 2021).

- Relative to our competitors, our sales growth was much better
- Relative to our competitors, our profit growth was much better
- Relative to our competitors, our market share was much better
- Relative to our competitors, our speed to market was much better
- Relative to our competitors, our net income was much better
- Relative to our competitors, our return on investment was much better

## Appendix 2

### Harman's Single-Factor

| Total Variance Explained |                     |               |              |                                     |               |              |
|--------------------------|---------------------|---------------|--------------|-------------------------------------|---------------|--------------|
| Component                | Initial Eigenvalues |               |              | Extraction Sums of Squared Loadings |               |              |
|                          | Total               | % of Variance | Cumulative % | Total                               | % of Variance | Cumulative % |
| 1                        | 15.404              | 39.498        | 39.498       | 15.404                              | 39.498        | 39.498       |
| 2                        | 2.811               | 7.209         | 46.707       |                                     |               |              |
| 3                        | 1.966               | 5.040         | 51.747       |                                     |               |              |
| 4                        | 1.747               | 4.480         | 56.227       |                                     |               |              |
| 5                        | 1.413               | 3.624         | 59.851       |                                     |               |              |
| 6                        | 1.326               | 3.399         | 63.250       |                                     |               |              |
| 7                        | 1.221               | 3.132         | 66.382       |                                     |               |              |
| 8                        | 1.051               | 2.695         | 69.077       |                                     |               |              |
| 9                        | .933                | 2.391         | 71.468       |                                     |               |              |
| 10                       | .815                | 2.090         | 73.558       |                                     |               |              |
| 11                       | .744                | 1.908         | 75.466       |                                     |               |              |
| 12                       | .676                | 1.734         | 77.200       |                                     |               |              |
| 13                       | .615                | 1.578         | 78.778       |                                     |               |              |
| 14                       | .583                | 1.494         | 80.271       |                                     |               |              |
| 15                       | .559                | 1.433         | 81.704       |                                     |               |              |
| 16                       | .526                | 1.349         | 83.054       |                                     |               |              |
| 17                       | .507                | 1.299         | 84.353       |                                     |               |              |
| 18                       | .482                | 1.235         | 85.588       |                                     |               |              |
| 19                       | .462                | 1.184         | 86.771       |                                     |               |              |
| 20                       | .452                | 1.159         | 87.930       |                                     |               |              |

(continued on next page)

**Appendix 2 (continued)**

| Total Variance Explained |                     |               |              |                                     |               |              |
|--------------------------|---------------------|---------------|--------------|-------------------------------------|---------------|--------------|
| Component                | Initial Eigenvalues |               |              | Extraction Sums of Squared Loadings |               |              |
|                          | Total               | % of Variance | Cumulative % | Total                               | % of Variance | Cumulative % |
| 21                       | .416                | 1.066         | 88.996       |                                     |               |              |
| 22                       | .399                | 1.023         | 90.019       |                                     |               |              |
| 23                       | .357                | .914          | 90.933       |                                     |               |              |
| 24                       | .331                | .849          | 91.783       |                                     |               |              |
| 25                       | .321                | .822          | 92.605       |                                     |               |              |
| 26                       | .295                | .756          | 93.360       |                                     |               |              |
| 27                       | .281                | .720          | 94.080       |                                     |               |              |
| 28                       | .270                | .691          | 94.771       |                                     |               |              |
| 29                       | .244                | .626          | 95.397       |                                     |               |              |
| 30                       | .230                | .591          | 95.988       |                                     |               |              |
| 31                       | .221                | .567          | 96.555       |                                     |               |              |
| 32                       | .201                | .515          | 97.070       |                                     |               |              |
| 33                       | .198                | .506          | 97.576       |                                     |               |              |
| 34                       | .185                | .474          | 98.050       |                                     |               |              |
| 35                       | .172                | .442          | 98.492       |                                     |               |              |
| 36                       | .169                | .434          | 98.926       |                                     |               |              |
| 37                       | .152                | .389          | 99.315       |                                     |               |              |
| 38                       | .140                | .358          | 99.673       |                                     |               |              |
| 39                       | .128                | .327          | 100.000      |                                     |               |              |

Extraction Method: Principal Component Analysis.

**Appendix 3****Normality Test**

| Tests of Normality |                                 |     |      |              |     |      |
|--------------------|---------------------------------|-----|------|--------------|-----|------|
|                    | Kolmogorov-Smirnov <sup>a</sup> |     |      | Shapiro-Wilk |     |      |
|                    | Statistic                       | df  | Sig. | Statistic    | df  | Sig. |
| New Capability     | .147                            | 264 | .000 | .943         | 264 | .000 |
| New Technology     | .116                            | 264 | .000 | .964         | 264 | .000 |
| New Partnership    | .122                            | 264 | .000 | .961         | 264 | .000 |
| New Process        | .195                            | 264 | .000 | .940         | 264 | .000 |
| New Offering       | .142                            | 264 | .000 | .962         | 264 | .000 |
| New Marketing      | .126                            | 264 | .000 | .957         | 264 | .000 |
| New Channel        | .138                            | 264 | .000 | .967         | 264 | .000 |
| New Relation       | .143                            | 264 | .000 | .960         | 264 | .000 |
| New Revenue        | .121                            | 264 | .000 | .971         | 264 | .000 |
| New Costing        | .109                            | 264 | .000 | .961         | 264 | .000 |
| Performance        | .071                            | 264 | .003 | .989         | 264 | .042 |

a. Lilliefors Significance Correction.

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